

Electric Current and DC Circuits

Charges in motion: current, resistance, power, and the rules for analysing every circuit from a torch to a pacemaker.

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So far our charges have been at rest. When a potential difference drives charges through a conductor, we get an **electric current**, the basis of all electrical technology. This chapter starts with a microscopic picture of current, using drift velocity and resistivity. It then builds up the circuit tools engineers use every day: Ohm's law, power, EMF and internal resistance, series and parallel combinations, Kirchhoff's rules, measuring instruments, and the charging and discharging of capacitors in RC circuits. We finish with the effects of current on the human body, the foundation of electrical safety in hospitals and homes.

YOU WILL BE ABLE TO

- Relate current to drift velocity and charge carrier density, and use current density.
- Use resistance, resistivity and its temperature dependence, and Ohm's law. Understand the microscopic (Drude) model of conduction.
- Calculate electrical power and energy, including Joule heating.
- Model real sources with an EMF and an internal resistance. Find terminal voltage and the condition for maximum power transfer.
- Reduce series-parallel networks, and solve multi-loop circuits with Kirchhoff's rules.
- Analyse ammeters, voltmeters and the Wheatstone bridge.
- Solve RC charging and discharging problems using the time constant.
- Explain the physiological effects of current and the principles of electrical safety.

THINK FIRST

Electrons drift through a household wire at well under a millimetre per second. Yet when you flip a switch, a light several metres away comes on almost instantly. How?

The wire is already *full* of free electrons. Closing the switch sets up an electric field along the whole circuit almost instantly, at close to the speed of light. Every electron everywhere in the circuit starts drifting at essentially the same moment. It is like a water pipe that is already full: open the tap and water flows out immediately, even though any particular water molecule moves slowly. The *signal* travels fast, while the charge carriers creep (Example 17.1).

§17.1

Electric current

Electric current is the rate at which charge flows through a cross-section:

$$I = \frac{dQ}{dt}.$$

The SI unit is the **ampere**: $1 \text{ A} = 1 \text{ C/s}$. By convention, current flows in the direction positive charges would move. In metals the carriers are actually electrons, which move the opposite way.

Drift velocity. Free electrons in a metal move randomly at very high speeds, about 10^6 m/s , colliding constantly with the vibrating ions. With no field, their average velocity is zero. A field gives them a small average **drift velocity** v_d opposite to $\vec{E}$. Suppose there are n carriers per unit volume, each with charge q , in a wire of cross-section A . In a time dt , all the carriers within a length $v_d dt$ pass through a cross-section, so $dQ = nqAv_d dt$ and

$$I = nqv_d A.$$

The **current density** is the current per unit area, $J = I/A = nqv_d$, measured in A/m^2 .

WORKED EXAMPLE 17.1

How slowly do electrons drift?

A copper wire of diameter 2.0 mm carries 10 A. Copper has about 8.5×10^{28} free electrons per m^3 . Find the drift velocity.

Solution

$$A = \pi(1.0 \times 10^{-3})^2 = 3.14 \times 10^{-6} \text{ m}^2.$$

$$v_d = \frac{I}{neA} = \frac{10}{(8.5 \times 10^{28})(1.602 \times 10^{-19})(3.14 \times 10^{-6})} = 2.3 \times 10^{-4} \text{ m/s}.$$

That is about 0.8 metres per hour.

Evaluate. The electrons in a car's battery cable take hours to travel its length. The energy, however, travels in the electromagnetic field around the wires at nearly the speed of light.

§17.2

Resistance, resistivity and Ohm's law

The **resistance** of a component is

$$R = \frac{V}{I},$$

measured in **ohms** ($1 \Omega = 1 \text{ V/A}$). A material obeys **Ohm's law** if R stays constant as V varies. Metals and resistors are close to ohmic. Diodes, transistors, filament lamps (as they heat up) and nerve membranes are not.

For a uniform wire of length L and cross-section A :

$$R = \rho \frac{L}{A},$$

where ρ is the **resistivity** of the material, in $\Omega \text{ m}$. Its inverse, $\sigma = 1/\rho$, is the **conductivity**. In microscopic form, Ohm's law reads $\vec{J} = \sigma \vec{E}$.

MATERIAL	ρ AT 20 °C ($\Omega \text{ m}$)
Silver	1.6×10^{-8}
Copper	1.7×10^{-8}
Aluminium	2.8×10^{-8}
Nichrome (heating elements)	1.1×10^{-6}
Seawater	0.2

MATERIAL	ρ AT 20 °C (Ω M)
Blood	about 1.5
Muscle tissue	2–5
Fat	20–30
Pure silicon	about 10^3
Glass	10^{10} – 10^{14}

Temperature dependence. For metals, the resistivity rises roughly linearly with temperature:

$$\rho(T) \approx \rho_0 [1 + \alpha(T - T_0)],$$

with $\alpha \approx 0.004 \text{ K}^{-1}$ for copper. Hotter ions vibrate more and scatter the electrons more often. Platinum resistance thermometers use this effect. Semiconductors behave the opposite way: their resistance *falls* as they warm, which thermistors exploit. Superconductors lose all resistance below a critical temperature, which lets MRI magnets carry hundreds of amperes with no heating.

The microscopic picture (Drude model). Between collisions, which happen on average every τ seconds, an electron accelerates at eE/m . Its average drift velocity is therefore $v_d = eE\tau/m$. Combining this with $J = nev_d$ gives

$$J = \frac{ne^2\tau}{m} E \implies \rho = \frac{m}{ne^2\tau}.$$

For copper, $\tau \approx 2.5 \times 10^{-14} \text{ s}$ (Problem P17.13).

§17.3

Electrical power

When a current I flows through a potential difference V , charge $dQ = I dt$ loses potential energy $V dQ$. So the power transferred is

$$P = IV.$$

For a resistor this energy becomes heat (**Joule heating**):

$$P = I^2 R = \frac{V^2}{R}.$$

Electricity suppliers charge for energy in kilowatt-hours: $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

WORKED EXAMPLE 17.2

An electric kettle

A 2.0 kW kettle runs on 230 V mains for 3.0 minutes. Find the current, the element's resistance, and the energy used in kWh.

Solution

$$I = \frac{P}{V} = \frac{2000}{230} = 8.7 \text{ A}, \quad R = \frac{V^2}{P} = \frac{230^2}{2000} = 26 \Omega.$$

$$E = Pt = 2.0 \text{ kW} \times 0.050 \text{ h} = 0.10 \text{ kWh} \quad (3.6 \times 10^5 \text{ J}).$$

Evaluate. Power is transmitted over long distances at very high voltage (hundreds of kV) because, for a given power, a higher voltage means a smaller current. The losses in the line, $I^2 R_{\text{line}}$, then fall with the *square* of the current.

§17.4

EMF and internal resistance

A battery or generator maintains a potential difference by doing non-electrostatic work on charges: chemical work in a battery, magnetic work in a generator. This work per unit charge is the **electromotive force** (EMF), $\mathcal{E}$, measured in volts. A real source also has an **internal resistance** r , so the voltage at its terminals when it delivers a current I is

$$V_{\text{terminal}} = \mathcal{E} - Ir.$$

Connected to an external load R :

$$I = \frac{\mathcal{E}}{R + r}.$$

Maximum power transfer. The power delivered to the load is $P = I^2 R = \mathcal{E}^2 R / (R + r)^2$. It is greatest when $R = r$ (Problem P17.14), at which point $P_{\text{max}} = \mathcal{E}^2 / 4r$ and only half of the source's power reaches the load.

WORKED EXAMPLE 17.3**A car battery under load**

A 12.0 V battery with an internal resistance of $0.50\ \Omega$ powers a $5.5\ \Omega$ load. Find the current, the terminal voltage, the power delivered to the load, and the power wasted inside the battery. What load would receive the most power?

Solution

$$I = \frac{12.0}{5.5 + 0.50} = 2.0\ \text{A}, \quad V_{\text{terminal}} = 12.0 - 2.0(0.50) = 11.0\ \text{V}.$$

$$P_{\text{load}} = I^2 R = 22\ \text{W}, \quad P_{\text{internal}} = I^2 r = 2.0\ \text{W}.$$

Maximum power transfer needs $R = r = 0.50\ \Omega$, giving $P = 12^2 / (4 \times 0.50) = 72\ \text{W}$.

Evaluate. When a car's starter motor draws 200 A, the internal voltage drop is large, which is why the headlights dim while the engine is cranking. A battery with a higher internal resistance (an old battery, or a cold one) may not be able to start the engine at all.

§17.5

Resistors in series and parallel**KEY IDEA**

Series (the same current flows through each, and the voltages add):

$$R_{\text{eq}} = R_1 + R_2 + \cdots$$

Parallel (the same voltage is across each, and the currents add):

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$$

For two resistors in parallel, $R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$. The equivalent resistance of a parallel combination is always smaller than the smallest resistor in it.

Household circuits are wired in **parallel**, so that every appliance receives the full mains voltage and can be switched on and off independently. Christmas lights wired in series all go out when one bulb fails, which is why modern sets use shunts or parallel wiring.

WORKED EXAMPLE 17.4**A series–parallel circuit**

A $4.0\ \Omega$ resistor is connected in series with a parallel pair of $6.0\ \Omega$ and $3.0\ \Omega$ resistors, across a $12\ \text{V}$ battery with negligible internal resistance. Find the current in each resistor.

Solution

Equivalent resistance:

$$R_{\parallel} = \frac{6.0 \times 3.0}{9.0} = 2.0\ \Omega, \quad R_{\text{eq}} = 4.0 + 2.0 = 6.0\ \Omega.$$

Currents:

- Total current, through the $4.0\ \Omega$ resistor: $I = 12/6.0 = 2.0\ \text{A}$.
- Voltage across the parallel pair: $V_{\parallel} = 2.0 \times 2.0 = 4.0\ \text{V}$.
- So $I_6 = 4.0/6.0 = 0.67\ \text{A}$ and $I_3 = 4.0/3.0 = 1.33\ \text{A}$. These add to $2.0\ \text{A}$. ✓

The smaller resistor takes the larger share of the current.

§17.6

Kirchhoff's rules

Many circuits cannot be reduced to series and parallel combinations. For these we use two rules that express conservation laws.

KEY IDEA

Junction rule (conservation of charge). The total current flowing into any junction equals the total current flowing out:

$$\sum I_{\text{in}} = \sum I_{\text{out}}.$$

Loop rule (conservation of energy). Around any closed loop, the potential differences add to zero:

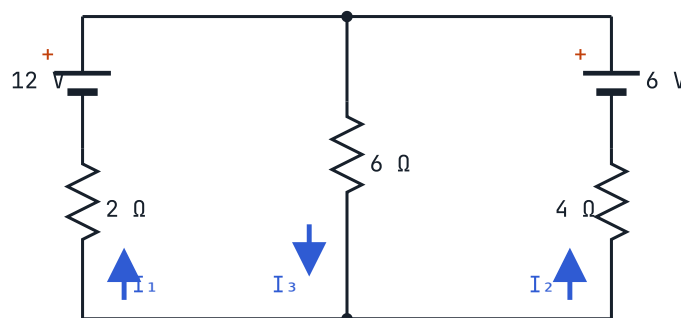
$$\sum \Delta V = 0.$$

Sign conventions for going round a loop:

- Crossing a resistor in the direction of the assumed current, $\Delta V = -IR$. Against the current, $+IR$.
- Crossing a battery from $-$ to $+$, $\Delta V = +\mathcal{E}$. From $+$ to $-$, $-\mathcal{E}$.

Method.

1. Label a current in every branch, guessing its direction.
2. Write junction equations, one fewer than the number of junctions.
3. Write loop equations until you have as many independent equations as unknown currents.
4. Solve. A negative answer simply means that the current flows opposite to the direction you guessed.



A two-loop circuit that cannot be reduced to series and parallel combinations. The arrows show the assumed current directions used in Example 17.5.

WORKED EXAMPLE 17.5**A two-loop circuit**

For the circuit in the figure, find the three currents.

Solution

Junction rule (top junction): $I_1 + I_2 = I_3$.

Left loop (clockwise from the bottom left: up through the 12 V battery, down through the 6 Ω resistor, and back through the 2 Ω resistor):

$$12 - 6I_3 - 2I_1 = 0.$$

Right loop (anticlockwise from the bottom right: up through the 6 V battery, down through the 6 Ω resistor, and back through the 4 Ω resistor):

$$6 - 6I_3 - 4I_2 = 0.$$

Solve. Substitute $I_3 = I_1 + I_2$:

$$8I_1 + 6I_2 = 12, \quad 6I_1 + 10I_2 = 6.$$

From the first equation, $I_1 = 1.5 - 0.75I_2$. Substituting into the second: $9 + 5.5I_2 = 6$, so

$$I_2 = -0.55 \text{ A}, \quad I_1 = 1.91 \text{ A}, \quad I_3 = 1.36 \text{ A}.$$

Interpretation. I_2 is negative, so the current in the right branch actually flows *downward* through the 6 V battery, from its + terminal to its – terminal. The 12 V battery is **charging** the 6 V battery.

Check (left loop): $12 - 6(1.36) - 2(1.91) = 12 - 8.18 - 3.82 = 0. \checkmark$

§17.7

Measuring instruments

- An **ammeter** is connected in **series** and must have a very *low* resistance, so that it does not disturb the current it measures.
- A **voltmeter** is connected in **parallel** and must have a very *high* resistance, so that it draws negligible current.

A moving-coil galvanometer (resistance R_g , full-scale current I_g) can be converted into either instrument:

- an **ammeter**, by adding a small **shunt** resistance in parallel, $R_{sh} = \frac{I_g R_g}{I - I_g}$;
- a **voltmeter**, by adding a large resistance in series, $R_s = \frac{V}{I_g} - R_g$.

The Wheatstone bridge. Four resistors are arranged in a diamond, with a galvanometer across the middle. When the bridge is *balanced*, no current flows through the galvanometer, and

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}.$$

Measuring at balance, where the current is zero, makes the result independent of the meter's own characteristics. Bridges are used to read **strain gauges**, which measure tiny changes in resistance when a structure flexes (Problem P17.12), and in many sensor circuits.

WORKED EXAMPLE 17.6

Building meters from a galvanometer

A galvanometer has a resistance of $50\ \Omega$ and a full-scale deflection at $1.0\ \text{mA}$. How would you convert it into (a) an ammeter reading up to $1.0\ \text{A}$ and (b) a voltmeter reading up to $10\ \text{V}$?

Solution

(a) **Ammeter:** a shunt in parallel carrying the other $0.999\ \text{A}$:

$$R_{sh} = \frac{(1.0 \times 10^{-3})(50)}{0.999} = 0.050\ \Omega.$$

The ammeter's total resistance is about $0.05\ \Omega$.

(b) **Voltmeter:** a resistance in series:

$$R_s = \frac{10}{1.0 \times 10^{-3}} - 50 = 9950\ \Omega.$$

The voltmeter's total resistance is $10\ \text{k}\Omega$, or “ $1000\ \Omega$ per volt”. Modern digital voltmeters have about $10\ \text{M}\Omega$.

§17.8

RC circuits

When a capacitor charges or discharges through a resistor, the current is not steady: it decays exponentially.

Charging. At $t = 0$, a switch connects an uncharged capacitor C in series with a resistor R to a battery of EMF $\mathcal{E}$. Going round the loop:

$$\mathcal{E} - iR - \frac{q}{C} = 0, \quad i = \frac{dq}{dt}.$$

Separating the variables and integrating, with $q(0) = 0$:

$$q(t) = C\mathcal{E} \left(1 - e^{-t/RC}\right), \quad i(t) = \frac{\mathcal{E}}{R} e^{-t/RC}.$$

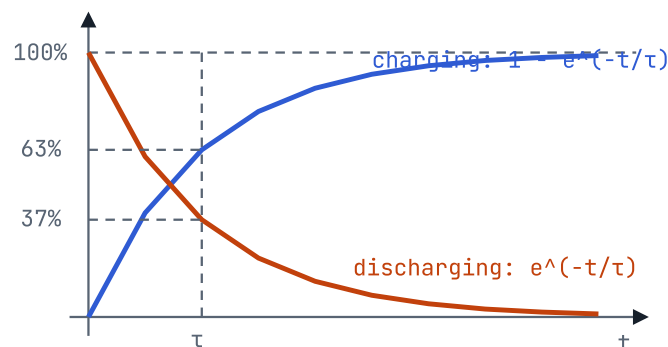
Discharging. A capacitor initially charged to q_0 discharges through R :

$$q(t) = q_0 e^{-t/RC}, \quad i(t) = -\frac{q_0}{RC} e^{-t/RC}.$$

KEY IDEA

Time constant: $\tau = RC$ (ohms $\times$ farads = seconds).

- After one time constant, a charging capacitor reaches $1 - e^{-1} = 63\%$ of its final charge, and a discharging one falls to $e^{-1} = 37\%$.
- After 5τ , the process is more than 99% complete.
- The time to fall to half is $\tau \ln 2 = 0.693\tau$.



Charging and discharging a capacitor. Each time constant $\tau = RC$ closes 63% of the remaining gap.

WORKED EXAMPLE 17.7

Charging a capacitor

A $10\ \mu\text{F}$ capacitor charges through a $100\ \text{k}\Omega$ resistor from a $12\ \text{V}$ battery. Find the time constant, the voltage across the capacitor after $2.0\ \text{s}$, the initial current, and the time to reach 99% of full charge.

Solution

$$\tau = RC = (1.0 \times 10^5)(1.0 \times 10^{-5}) = 1.0\ \text{s}.$$

- After $2.0\ \text{s}$: $V_C = 12(1 - e^{-2}) = 12(0.865) = 10.4\ \text{V}$.
- Initial current: $i_0 = \mathcal{E}/R = 12/10^5 = 0.12\ \text{mA}$.
- 99% charged when $e^{-t/\tau} = 0.01$, that is, $t = \tau \ln 100 = 4.6\ \text{s}$.

IN PRACTICE

RC timing in medicine and electronics. RC circuits set the timing of everything from windscreen-wiper delays and camera shutters to the oscillators in classic pacemaker designs. In those pacemakers, a capacitor charges through a resistor until it reaches a threshold, then fires a pulse to the heart and resets, so that RC sets the heart rate. Cell membranes behave like RC circuits too: their resistance and capacitance give a *membrane time constant* of a few to tens of milliseconds (Problem P17.17). That time constant controls how a neuron adds up its many incoming signals.

§17.9

Current and the human body

Current, not voltage, causes injury. The voltage matters only because, with the body's resistance, it determines the current. Typical effects of 50–60 Hz alternating current passing through the trunk for about a second are:

CURRENT	EFFECT
about 1 mA	threshold of perception (tingling)
about 5 mA	maximum “harmless” current

CURRENT	EFFECT
10–20 mA	muscles contract involuntarily: you cannot let go
50–100 mA	pain, breathing difficulty, risk of ventricular fibrillation
> 1 A	severe burns, cardiac arrest, tissue damage

The body's resistance is dominated by the skin. Dry skin may give 100 kΩ or more from hand to hand, but wet or broken skin can drop the total to about 1 kΩ. Internally, the body is a salty conductor of only a few hundred ohms. In hospitals, the greatest danger is **microshock**: a catheter or pacing lead that reaches the heart bypasses the skin's resistance completely. Then currents as small as 10–100 μA, far too small to feel, can trigger fibrillation. Medical equipment is therefore built and tested to strict leakage-current limits.

WORKED EXAMPLE 17.8

Wet hands and mains voltage

A person touches a 230 V live conductor with one hand while the other hand rests on an earthed metal tap. Estimate the current if the total path resistance is (a) 100 kΩ (dry skin) and (b) 1.0 kΩ (wet skin). How does a residual-current device (RCD) protect them?

Solution

(a) $I = 230/10^5 = 2.3$ mA: an unpleasant tingle.

(b) $I = 230/10^3 = 230$ mA: likely ventricular fibrillation, and potentially fatal.

Protection. An RCD (also called a GFCI) continuously compares the current in the live and neutral wires. Normally they are equal. If some current escapes to earth through a person, they differ, and the RCD cuts the supply when the difference exceeds about 30 mA, within roughly 30 ms. That is fast enough, in most cases, to prevent fibrillation. **Earthing** metal cases ensures that a fault produces a large current to earth, which blows the fuse rather than leaving the case “live” for someone to touch.

CHAPTER SUMMARY

- $I = dQ/dt = nqv_d A$. Drift speeds are of order mm/s, while signals travel at nearly c .
- $R = V/I$ and $R = \rho L/A$. Metals: ρ rises with T . Drude model: $\rho = m/(ne^2\tau)$.
- $P = IV = I^2 R = V^2/R$. $1 \text{ kWh} = 3.6 \text{ MJ}$.
- Real source: $V = \mathcal{E} - Ir$ and $I = \mathcal{E}/(R + r)$. Maximum power transfer when $R = r$.
- Series: $R_{\text{eq}} = \sum R_i$. Parallel: $1/R_{\text{eq}} = \sum 1/R_i$.
- Kirchhoff: $\sum I_{\text{in}} = \sum I_{\text{out}}$ at a junction, and $\sum \Delta V = 0$ around a loop.
- Ammeter: in series, low resistance (shunt). Voltmeter: in parallel, high resistance. Balanced bridge: $R_1/R_2 = R_3/R_4$.
- RC circuits: $\tau = RC$. Charging: $q = C\mathcal{E}(1 - e^{-t/\tau})$. Discharging: $q = q_0 e^{-t/\tau}$.
- Safety: about 10 mA means “can’t let go”, and about 100 mA risks fibrillation. Skin resistance dominates. RCDs trip at about 30 mA.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P17.1

If electrons drift at less than a millimetre per second, why does a lamp light up almost instantly when the switch is closed?

P17.2

A 60 W bulb and a 100 W bulb (both rated for 230 V) are connected in series across the 230 V mains. Which glows more brightly? Explain.

P17.3

Why are household appliances connected in parallel rather than in series?

P17.4

When a car's starter motor runs, the headlights dim. Explain this using the idea of internal resistance.

LEVEL B – STANDARD PROBLEMS**P17.5**

Overhead power lines are made of aluminium rather than copper. Compare the cross-sectional areas and masses of aluminium and copper wires of the same length and resistance. (Resistivities: Al $2.8 \times 10^{-8} \Omega \text{ m}$, Cu $1.7 \times 10^{-8} \Omega \text{ m}$. Densities: Al 2700 kg/m^3 , Cu 8960 kg/m^3 .)

P17.6

A 1000 W, 230 V heater uses nichrome wire ($\rho = 1.1 \times 10^{-6} \Omega \text{ m}$) of diameter 0.50 mm. Find the resistance of the element and the length of wire needed.

P17.7

A battery has an open-circuit voltage of 9.0 V. When it delivers 0.50 A to a 16Ω resistor, its terminal voltage falls to 8.0 V. Find its EMF and internal resistance, and its short-circuit current.

P17.8

A 6.0Ω and a 12Ω resistor are connected in parallel. This pair is in series with an 8.0Ω resistor, and that combination is in parallel with a 12Ω resistor, all across an 18 V battery. Find the equivalent resistance, the total current, and the current in the 8.0Ω resistor.

P17.9

A healthy car battery (EMF 12.6 V, internal resistance 0.020Ω) is connected by jumper cables (total resistance 0.010Ω) to a flat battery (EMF 10.0 V, internal resistance 0.050Ω), positive to positive. Find the charging current, and the power dissipated in each resistance.

P17.10

A camera flash charges a $1000 \mu\text{F}$ capacitor through a $2.0 \text{ k}\Omega$ resistor. Find the time constant, and the time needed to reach 90% of full charge.

P17.11

Electric shock. With the supply at 230 V, compare the current through a person whose hand-to-hand resistance is (a) 50 k Ω and (b) 1.0 k Ω . Use the table in the notes to describe the likely effects. Why is a hospital patient with a cardiac catheter at risk from currents of only tens of microamperes?

P17.12

Strain gauge. A Wheatstone bridge has four arms of 350 Ω and is excited by 10 V. One arm is a strain gauge whose resistance increases by 0.20% when the structure it is bonded to is stretched. Show that the bridge output is approximately $V\Delta R/(4R)$, and evaluate it.

LEVEL C – CHALLENGE PROBLEMS**P17.13**

Drude model. (a) Derive $\rho = m/(ne^2\tau)$ from the picture of electrons accelerating freely between collisions. (b) Find τ for copper. (c) Using the typical electron speed in copper of 1.6×10^6 m/s (the Fermi speed), find the mean free path between collisions, and compare it with the spacing between copper atoms (0.26 nm). What does this suggest about what the electrons are really colliding with?

P17.14

(a) Show that a source of EMF $\mathcal{E}$ and internal resistance r delivers maximum power to a load when $R = r$. (b) Show that the efficiency (load power $\div$ total power) is then only 50%. (c) Explain why power stations and power grids are deliberately *not* operated at the maximum-power condition, while audio amplifiers and radio antennas often are matched.

P17.15

An infinite ladder network consists of identical resistors R in each series link and in each shunt (rung). Show that the equivalent resistance between its input terminals is $R(1 + \sqrt{5})/2$.

P17.16

A capacitor C is charged from zero to a voltage $\mathcal{E}$ through a resistor R , using a battery of EMF $\mathcal{E}$. (a) By integrating $i^2 R$ over time, show that the energy dissipated in the resistor is $\frac{1}{2} C \mathcal{E}^2$, whatever the value of R . (b) Show that the battery supplies $C \mathcal{E}^2$ in total. What fraction ends up stored in the capacitor?

P17.17

Membrane time constant. A patch of nerve membrane has a specific resistance $R_m = 1.0 \, \Omega \, \text{m}^2$ (that is, a $1 \, \text{m}^2$ patch would have a resistance of $1.0 \, \Omega$ across it) and a specific capacitance $c_m = 0.010 \, \text{F/m}^2$. (a) Show that the membrane time constant $\tau = R_m c_m$ does not depend on the area of the patch, and evaluate it. (b) A brief current pulse raises the membrane voltage by $10 \, \text{mV}$ above rest. How long does it take for this disturbance to decay to $1 \, \text{mV}$? (c) Myelin wraps an axon in many layers of membrane, multiplying R_m by about 100 and dividing c_m by about 100. How does this change τ ? Why does myelin nevertheless speed up conduction?

P17.18

Bioimpedance. Body-composition analysers pass a tiny, safe alternating current through the body and measure its resistance R . Model the conducting tissue as a uniform cylinder of length H (the person's height) and volume V . (a) Show that $V = \rho H^2 / R$, where ρ is the effective resistivity of the conducting tissue. (b) For a $70 \, \text{kg}$ person $1.75 \, \text{m}$ tall with $R = 500 \, \Omega$, taking $\rho = 6.5 \, \Omega \, \text{m}$, estimate the volume of total body water. (c) Lean tissue is about 73% water, and fat contains very little. Estimate the fat-free mass and the body-fat percentage. Why does dehydration distort the measurement?

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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